Need to model *all* randomness with sigma points

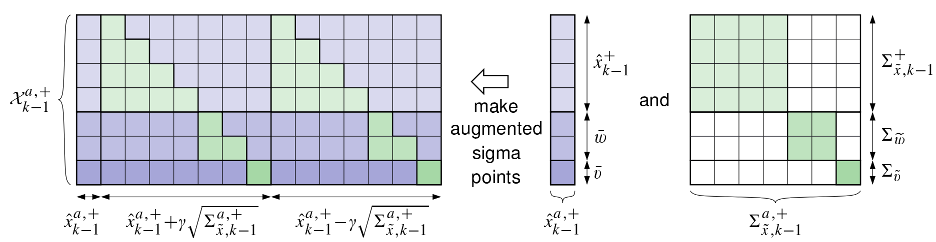
- We now apply sigma-point approach of propagating statistics through a nonlinear function to the state-estimation problem.
- These sigma-points must jointly model *all* randomness:
  - Uncertainty of the state.
  - Uncertainty of process noise.
  - Uncertainty of sensor noise.
- So we first define an augmented random vector x_k^a that combines these random factors at time index k.

Step 1a: State prediction time update (1)

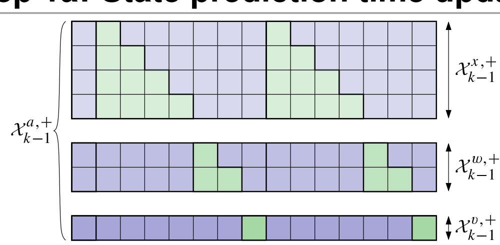
- First, form augmented prior state prediction, covariance:
  $\hat{x}_{k-1}^{a,+} = \big[(\hat{x}_{k-1}^{+})^T, \bar{w}^T, \bar{v}^T\big]^T$ and $\Sigma_{\tilde{x},k-1}^{a,+} = \operatorname{diag}\big(\Sigma_{\tilde{x},k-1}^{+}, \Sigma_{\widetilde{w}}, \Sigma_{\tilde{v}}\big)$.
- These factors are used to generate the $p+1$ augmented sigma points:

$$\mathcal{X}_{k-1}^{a,+} = \left\{ \hat{x}_{k-1}^{a,+},\ \hat{x}_{k-1}^{a,+} + \gamma\sqrt{\Sigma_{\tilde{x},k-1}^{a,+}},\ \hat{x}_{k-1}^{a,+} - \gamma\sqrt{\Sigma_{\tilde{x},k-1}^{a,+}} \right\}.$$

- Can be organized in convenient matrix form:

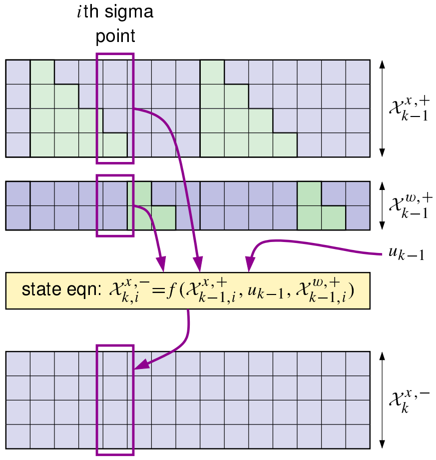


Step 1a: State prediction time update (2)



- Split augmented sigma points $\mathcal{X}_{k-1}^{a,+}$ into state portion $\mathcal{X}_{k-1}^{x,+}$, process-noise portion $\mathcal{X}_{k-1}^{w,+}$, and sensor-noise portion $\mathcal{X}_{k}^{v,+}$.

Step 1a: State prediction time update (3)

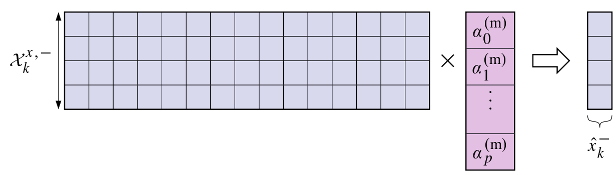


- Evaluate state equation using all pairs of $\mathcal{X}^{x,+}_{k-1,i}$ and $\mathcal{X}^{w,+}_{k-1,i}$ (where subscript i denotes that the ith vector is being extracted from the original set), yielding the prediction sigma points $\mathcal{X}^{x,-}_{k,i}$.
- That is, compute $\mathcal{X}^{x,-}_{k,i} = f(\mathcal{X}^{x,+}_{k-1,i}, u_{k-1}, \mathcal{X}^{w,+}_{k-1,i})$.

Step 1a–b: State prediction time update (4)

- Finally, present state prediction is computed as:

$$\begin{aligned}\hat{x}_k^- &= \mathbb{E}\big[f(x_{k-1}, u_{k-1}, w_{k-1}) \mid \mathbb{Z}_{k-1}\big] \\ &\approx \sum_{i=0}^{p} \alpha_i^{(\mathrm{m})} f(\mathcal{X}^{x,+}_{k-1,i}, u_{k-1}, \mathcal{X}^{w,+}_{k-1,i}) \\ &= \sum_{i=0}^{p} \alpha_i^{(\mathrm{m})} \mathcal{X}^{x,-}_{k,i} = \big[\mathcal{X}^{x,-}_{k}\big]\big[\alpha^{(\mathrm{m})}\big]\end{aligned}$$



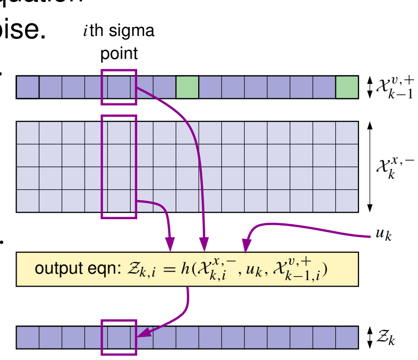
- Can compute with simple matrix multiply.
- Then, covariance estimate is computed as:

$$\begin{aligned}\Sigma^-_{\tilde{x},k} &= \textstyle\sum_{i=0}^{p} \alpha_i^{(\mathrm{c})} \big(\mathcal{X}^{x,-}_{k,i} - \hat{x}_k^-\big)\big(\mathcal{X}^{x,-}_{k,i} - \hat{x}_k^-\big)^T \\ &= \big[\mathcal{X}^{x,-}_{k} - \hat{x}_k^-\big]\big[\mathrm{diag}(\alpha^{(\mathrm{c})})\big]\big[\mathcal{X}^{x,-}_{k} - \hat{x}_k^-\big]^T\end{aligned}$$

Step 1c: Predict system output z_k

- Output z_k predicted by evaluating model output equation using sigma points describing state and sensor noise.
- First, compute points: $\mathcal{Z}_{k,i} = h(\mathcal{X}^{x,-}_{k,i}, u_k, \mathcal{X}^{v,+}_{k-1,i})$.
- Output estimate is then:

$$\begin{aligned}\hat{z}_k &= \mathbb{E}\big[h(x_k, u_k, v_k) \mid \mathbb{Z}_{k-1}\big] \\ &\approx \sum_{i=0}^{p} \alpha_i^{(\mathrm{m})} h(\mathcal{X}^{x,-}_{k,i}, u_k, \mathcal{X}^{v,+}_{k-1,i}) = \sum_{i=0}^{p} \alpha_i^{(\mathrm{m})} \mathcal{Z}_{k,i}.\end{aligned}$$



- Can be computed with simple matrix multiplication, as was done when calculating $\hat{x}_k^-$ at end of Step 1a.

Step 2a: Estimator gain matrix L_k

- To find L_k, must first compute required covariance matrices:

$$\Sigma_{\tilde{z},k} = \sum_{i=0}^{p} \alpha_i^{(c)} \big(\mathcal{Z}_{k,i} - \hat{z}_k\big)\big(\mathcal{Z}_{k,i} - \hat{z}_k\big)^T$$

$$\Sigma^-_{\tilde{x}\tilde{z},k} = \sum_{i=0}^{p} \alpha_i^{(c)} \big(\mathcal{X}^{x,-}_{k,i} - \hat{x}^-_k\big)\big(\mathcal{Z}_{k,i} - \hat{z}_k\big)^T.$$

- These depend on sigma-point matrices $\mathcal{X}^{x,-}_k$ and $\mathcal{Z}_k$, already computed in Steps 1b and 1c, as well as $\hat{x}^-_k$ and $\hat{y}_k$, already computed in Steps 1a and 1c.
- The summations can be performed using matrix multiplies, as we did in Step 1b.
- Then, we simply compute $L_k = \Sigma^-_{\tilde{x}\tilde{z},k}\Sigma^{-1}_{\tilde{z},k}$.

Step 2b–c: State, covariance measurement update

- Then the state estimate is computed as:

$$\hat{x}^+_k = \hat{x}^-_k + L_k(z_k - \hat{z}_k).$$

- Finally, the estimation-error covariance matrix is calculated directly from the optimal formulation:

$$\Sigma^+_{\tilde{x},k} = \Sigma^-_{\tilde{x},k} - L_k \Sigma_{\tilde{z},k} L_k^T.$$

Summary

- SPKF uses sigma-point method to propagate uncertainty of input RV to output of model's (possibly) nonlinear state and output equations.
- Applying this procedure to generic-probabilistic-inference solution yields all six SPKF steps.
- Matrices and vectors are convenient way to store all the sigma points and to compute means and covariances from the sigma points.
- SPKF is now derived!